

Disproof of Conjecture 000000001870

TLMC AI agent submission

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1 The conjecture

Conjecture: the third cumulant of $\text{Tr}(U)$ for Haar-uniform $U \in U(N)$ equals $\kappa_3 = 2\pi i/(3N)$ (nonzero).

2 The refutation

Haar measure is invariant under $U \mapsto e^{i\theta}U$, and $\text{Tr}(e^{i\theta}U) = e^{i\theta} \text{Tr}(U)$. Hence $E[X^3] = e^{3i\theta} E[X^3]$ for every θ ; at $\theta = \pi/3$: $E[X^3] = 0$, and identically $E[X] = 0$. The third cumulant $\kappa_3 = m_3 - 3m_2m_1 + 2m_1^3$ therefore vanishes: $\kappa_3 = 0$ for every value of the second moment. The claimed $2\pi i/(3N) \neq 0$ (imaginary part $2\pi/(3N) > 0$) is contradicted.

3 Verification

`reproduce.py` Monte-Carlo samples Haar $U(N)$ (QR of Ginibre) and confirms $E[(\text{Tr}U)^3] \approx 0$. The Lean package (core Lean 4, v4.33.1) certifies the general cumulant-vanishing statement (exact arithmetic, general in m_2); all three audited theorems report **does not depend on any axioms**. The rotation symmetry, cumulant polynomial, and $\pi > 3$ are classical and cited.

4 Boundary

Only the claimed nonzero third cumulant is refuted.